Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

Department of CSE

B.Tech. CSE Trimester -5

Roll Number-2520030384
Name-B. VR VR SrisaiHarini

Computer Networks and Network Programming
25CS2208E

LAB RECORD

2026

Experiment -1

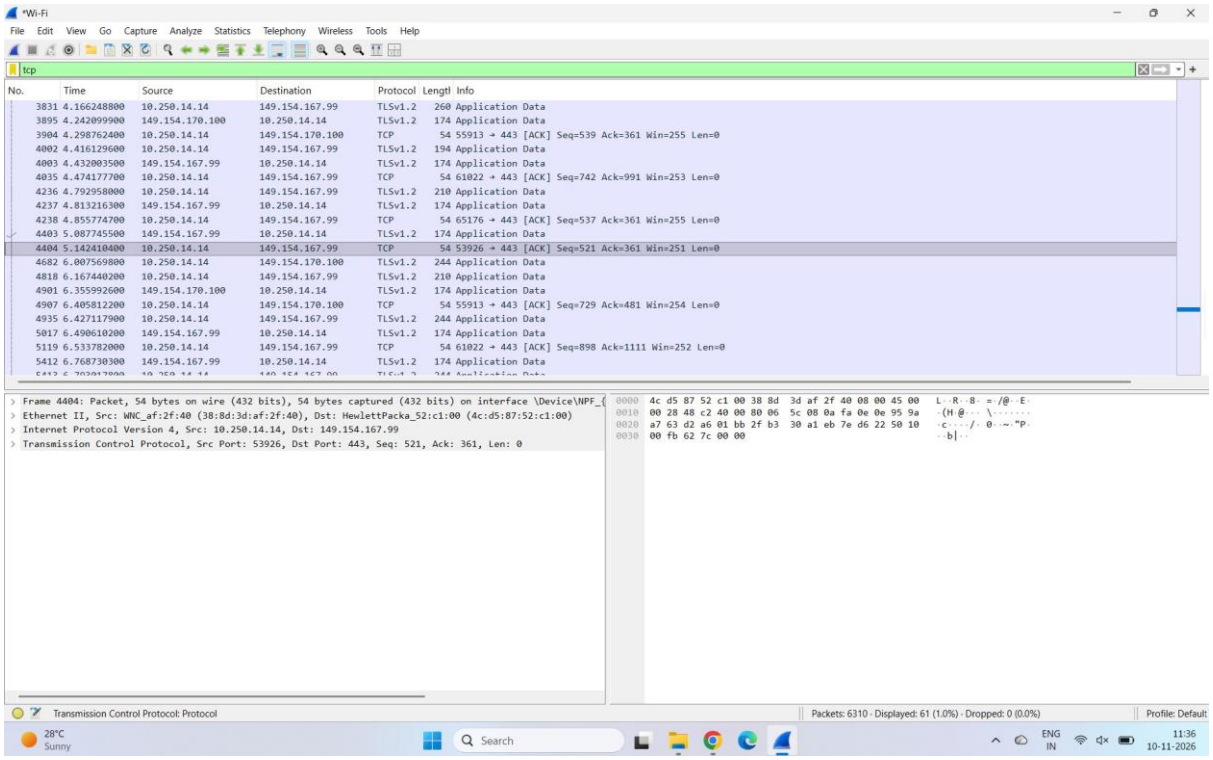
AIM: To analyse a PCAP (Packet Capture) file using Wireshark, identify the total number of captured packets, determine the number of TCP, UDP, and ICMP packets using display filters, and identify the top five protocols using the Protocol Hierarchy statistics.

Steps:

Screenshot 1: Filtering TCP Packets

Steps:

1. Open the captured PCAP file in Wireshark.
2. Enter tcp in the display filter bar.
3. Press Enter to apply the filter.
4. Wireshark displays only TCP-related packets.
5. Observe the source IP, destination IP, protocol, packet length and information columns.
6. Select a packet to view its Ethernet, IP and TCP headers in the packet details section.



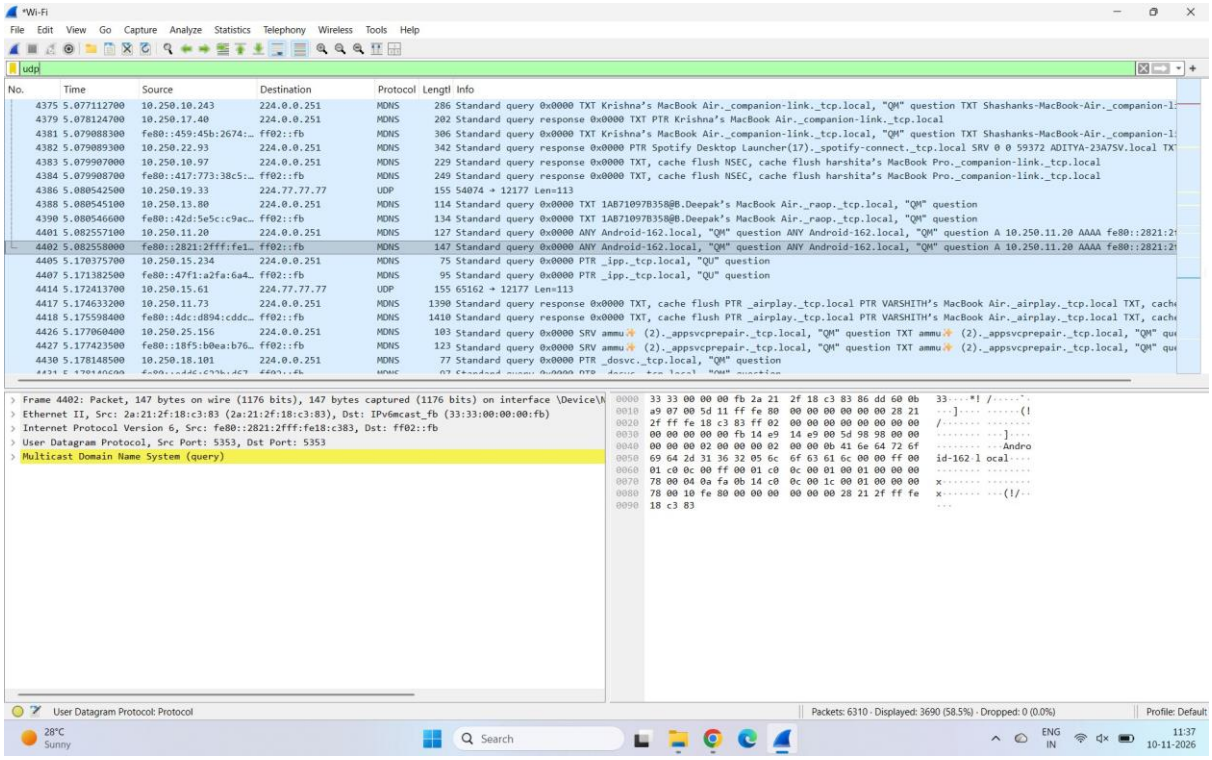
Explanation:

TCP (Transmission Control Protocol) is a reliable communication protocol that ensures data is delivered correctly.

Screenshot 2: Filtering UDP Packets

Steps:

1. Clear the previous filter from the display filter bar.
2. Enter udp in the filter bar.
3. Press Enter to apply the filter.
4. Observe the UDP packets displayed in the packet list.
5. Check the source and destination addresses, packet length and protocol information.
6. Select a packet to examine its Ethernet, IP and UDP headers.



Explanation:

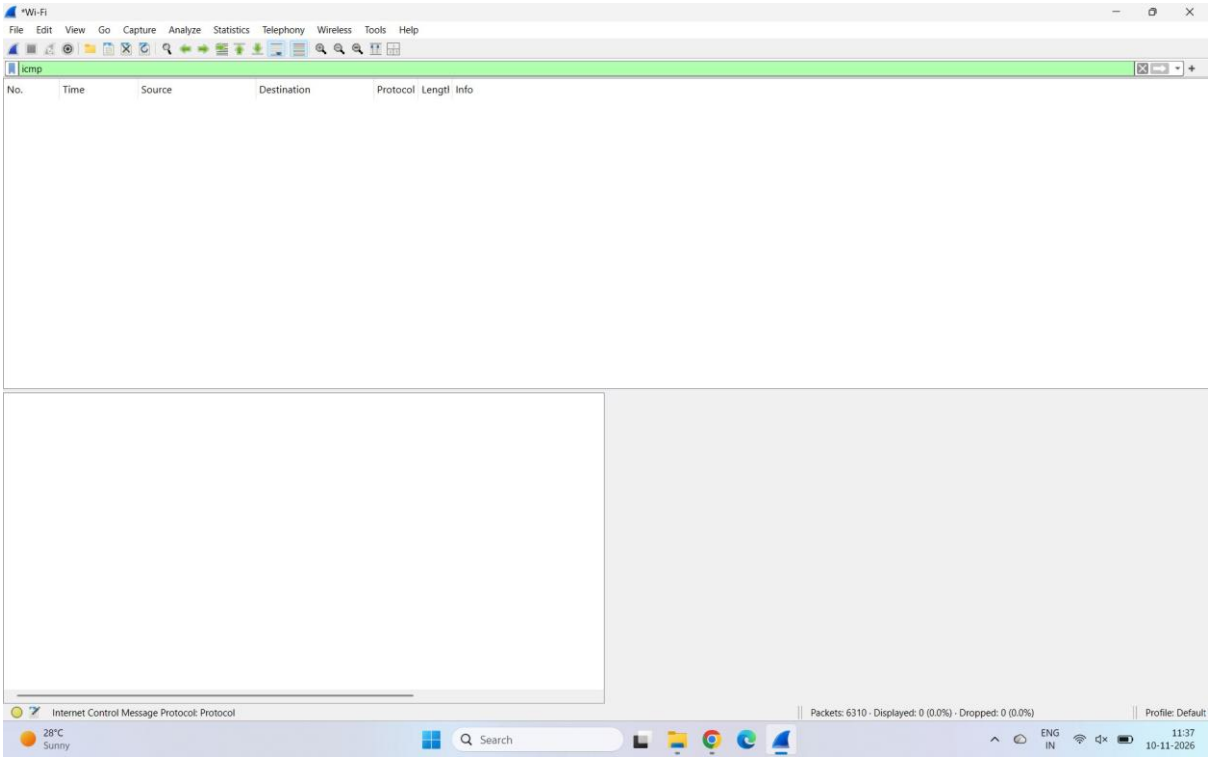
UDP (User Datagram Protocol) is a connectionless protocol that sends data without establishing a connection beforehand. In this screenshot, 3690 UDP

packets are displayed, representing approximately 58.5% of the total captured packets.

Screenshot 3: Filtering ICMP Packets

Steps:

1. Clear the previous display filter.
2. Enter icmp in the filter bar.
3. Press Enter to apply the filter.
4. Observe the packet list.
5. Check whether any ICMP packets are captured.



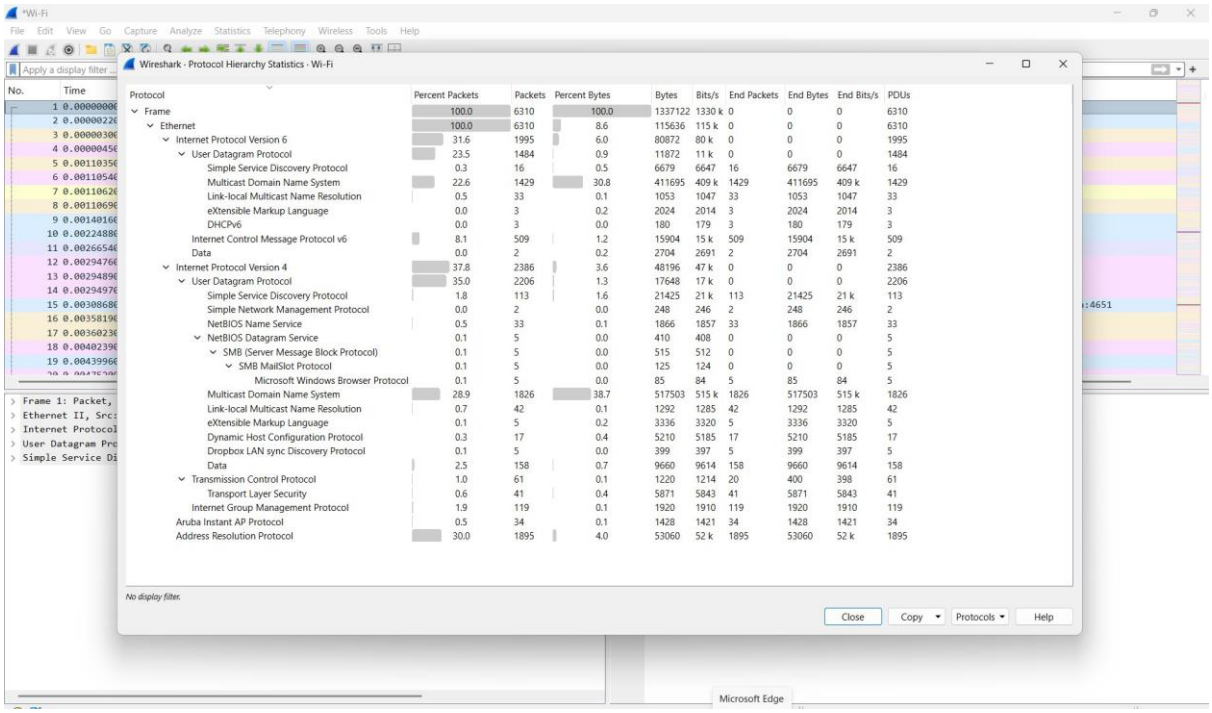
Explanation:

ICMP (Internet Control Message Protocol) is used for network error reporting and diagnostic operations, such as checking connectivity using the ping command. In this screenshot, no packets are displayed because there are zero IPv4 ICMP packets in the captured traffic.

Screenshot 4: Protocol Hierarchy Statistics

Steps:

1. Remove the display filter to view the complete packet capture.
2. Click on the Statistics menu in Wireshark.
3. Select Protocol Hierarchy.
4. A window displaying the protocol distribution appears.
5. Observe the number of packets and percentage occupied by each protocol.
6. Expand the protocol categories to examine Ethernet, IPv4, IPv6, TCP, UDP and other protocols.



Explanation:

Protocol Hierarchy provides an overall summary of the protocols present in the captured network traffic. It shows the number and percentage of packets belonging to each protocol.